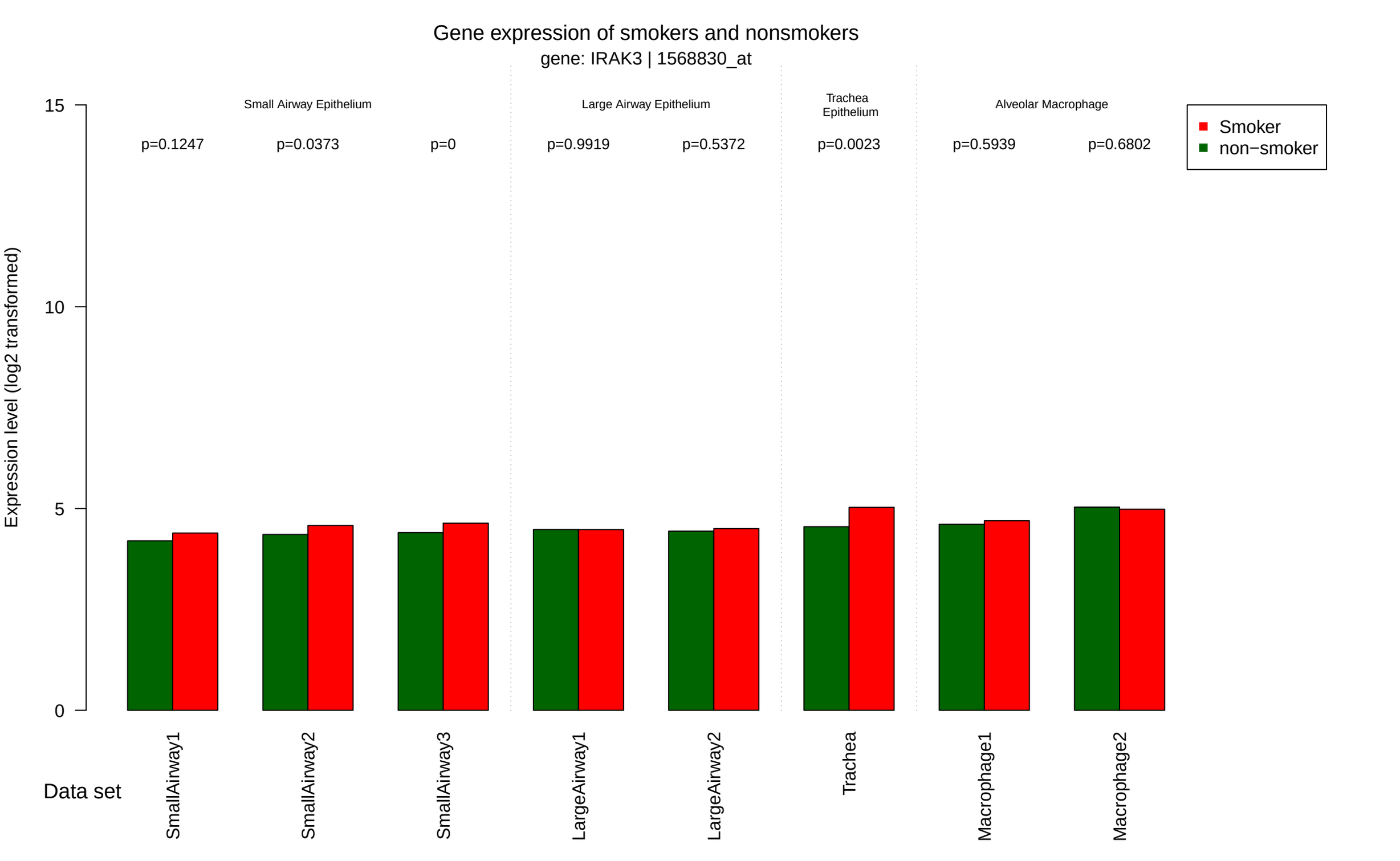
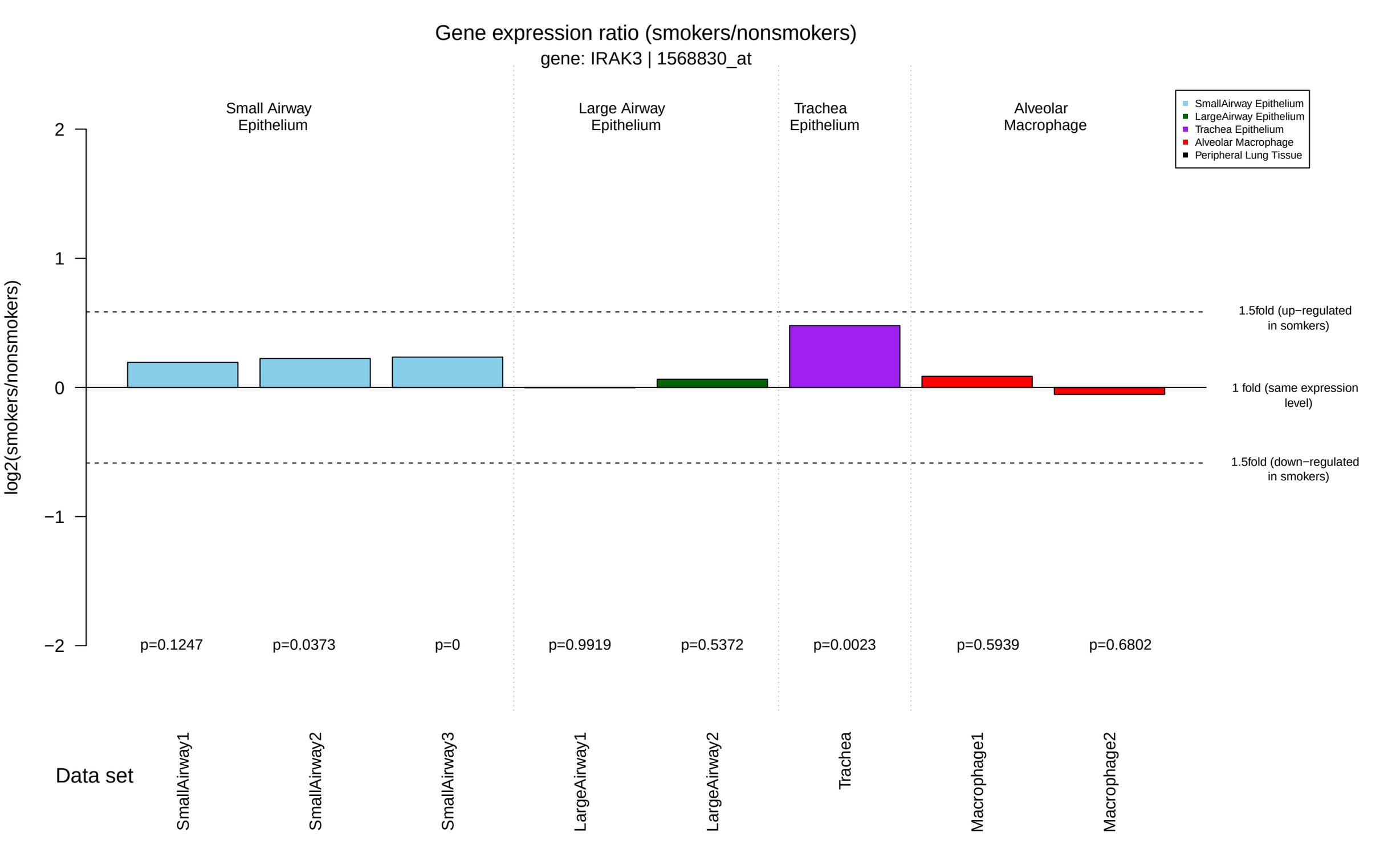
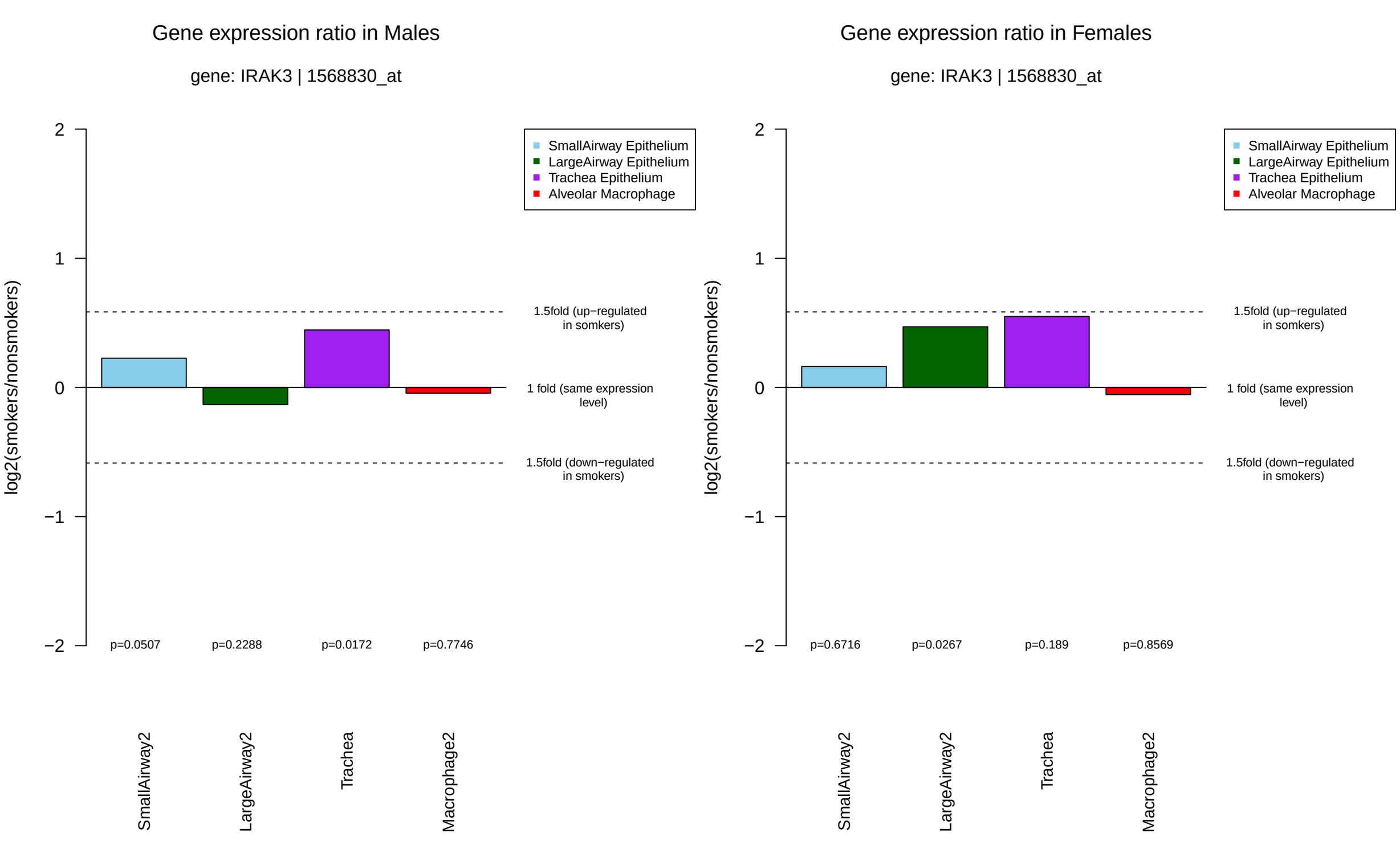
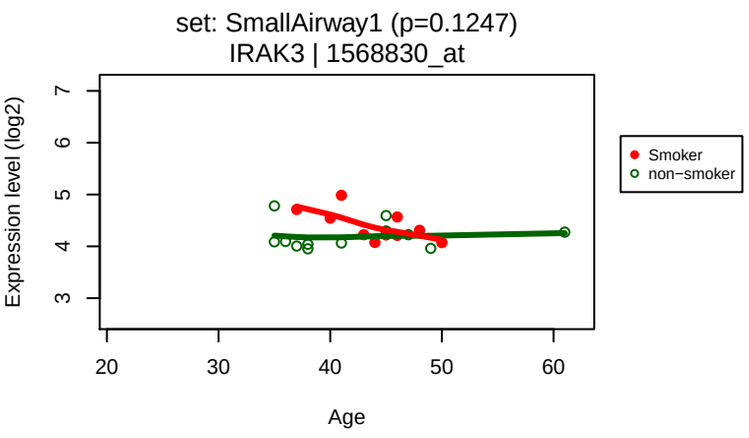
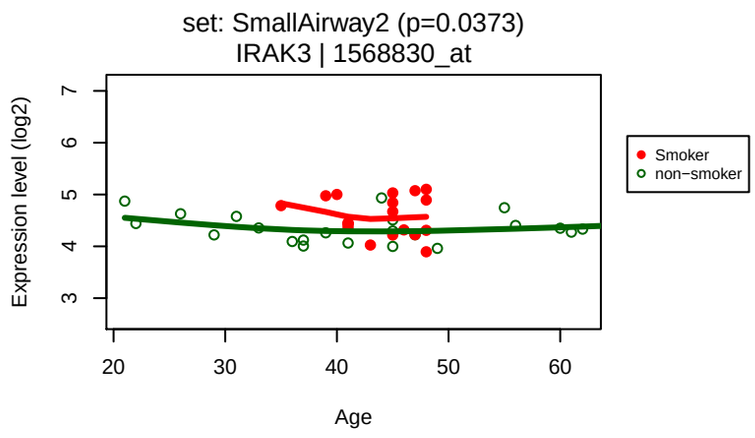
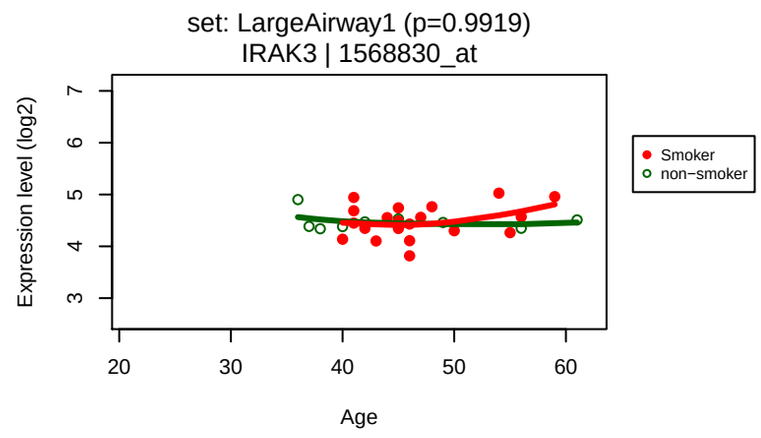

Gene expression of smokers and nonsmokers
gene: IRAK3 | 1568830_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1247
p=0.0373
p=0
p=0.9919
p=0.5372
p=0.0023
p=0.5939
p=0.6802
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

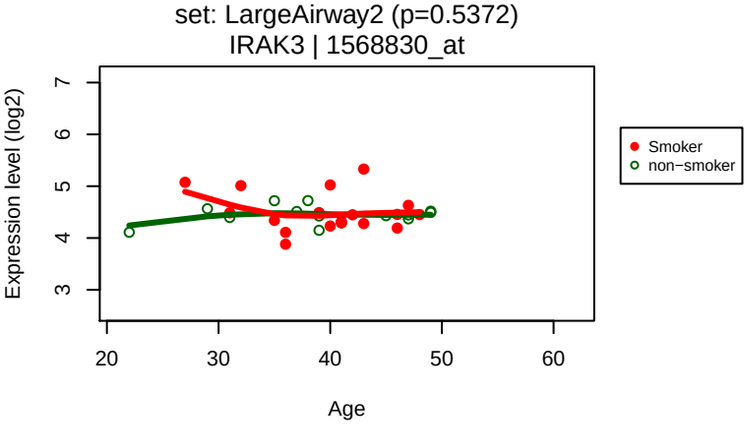

Gene expression ratio (smokers/nonsmokers)
gene: IRAK3 | 1568830_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1247
p=0.0373
p=0
p=0.9919
p=0.5372
p=0.0023
p=0.5939
p=0.6802
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

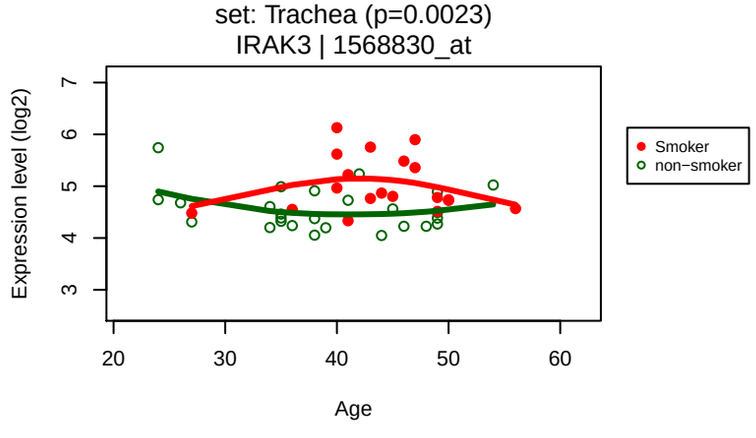

Gene expression ratio in Males
gene: IRAK3 | 1568830_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0507
p=0.2288
p=0.0172
p=0.7746
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IRAK3 | 1568830_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6716
p=0.0267
p=0.189
p=0.8569
SmallAirway2
LargeAirway2
Trachea
Macrophage2

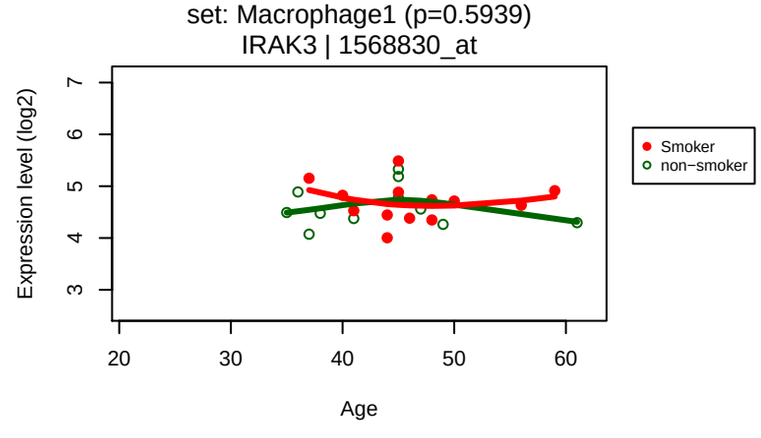

set: SmallAirway1 (p=0.1247)
IRAK3 | 1568830_at
Expression level (log2)
Age
Smoker
non−smoker

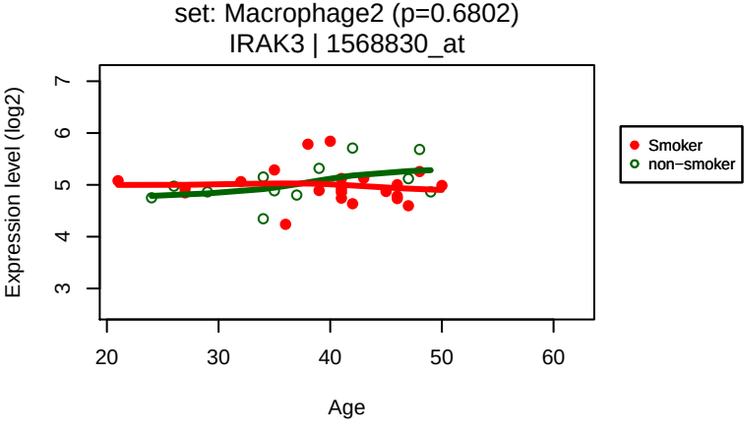

set: SmallAirway2 (p=0.0373)
IRAK3 | 1568830_at
Expression level (log2)
Age
Smoker
non−smoker

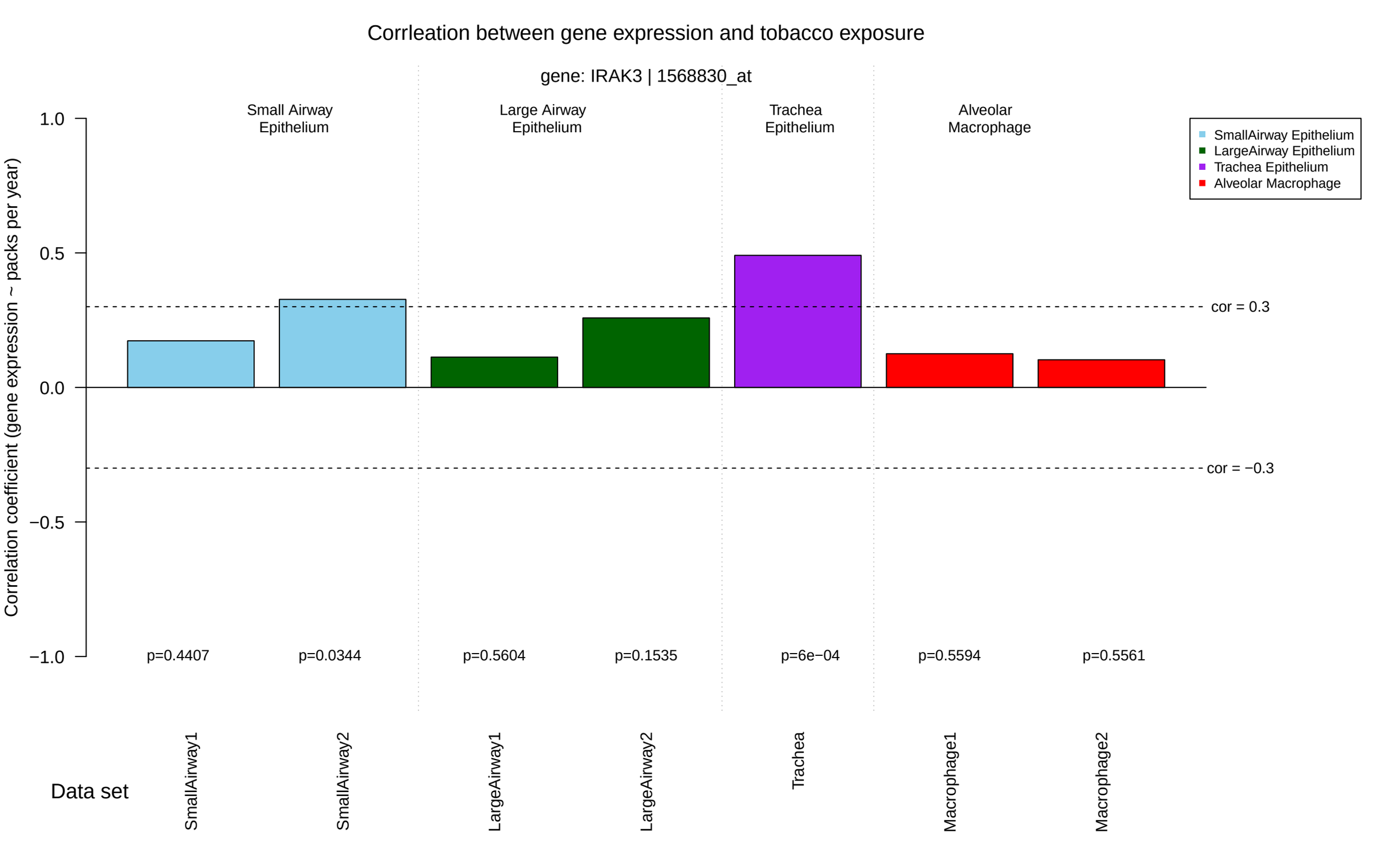

set: LargeAirway1 (p=0.9919)
IRAK3 | 1568830_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5372)
IRAK3 | 1568830_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0023)
IRAK3 | 1568830_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5939)
IRAK3 | 1568830_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6802)
IRAK3 | 1568830_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IRAK3 | 1568830_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4407
p=0.0344
p=0.5604
p=0.1535
p=6e−04
p=0.5594
p=0.5561
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2